

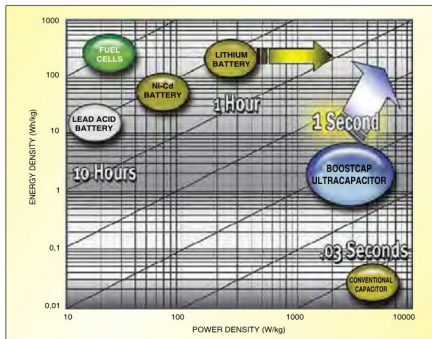


Fig. 1: Regon's plot showing energy density vs power density of various energy storage devices

Supercapacitors: Advantages & disadvantages

Advantages

1. High power and moderate energy density
2. Very fast charging and discharging
3. Very high capacitance
4. Linear charging and discharging (up to 0V)
5. State of charge present is a direct function of the output voltage
6. Very long shelf and working life (theoretically no limit!)
7. Direct short circuit is not fatal
8. Small over-charging is not a problem
9. Today, it is available up to 5000F, 220V
10. Series current-limiting resistor is not required as it accept charges as much as you supply

Disadvantages

1. Each cell can give 1V to 2.5V (limit due to electrolyte stability)
2. High cost
3. Very fast discharging
4. Low energy density
5. Peak power is limited by equivalent series resistance (ESR)
6. Time constant is small (1-2 second only)
7. Rapid discharge at severe rate increases the temperature
8. Complete discharge takes a few hours
9. Operating temperature is up to 65 degrees
10. It loses 2 per cent energy per day in self-discharge

Some leading manufacturers of supercapacitors

- NEC, Japan (for memory backups)
- Panasonic, Japan (for memory backups)
- Maxwell Technologies, USA
- EPCOS, Germany
- Evans Capacitor, USA (for automobiles)
- ELNA, USA (for automobiles)
- Cooper, USA (for automobiles)
- Kold Ban International, Australia
- CAP-XX, Australia
- Ness Capacitor, Korea
- Tavrima, Canada
- ESMA, Russia (for power quality applications)

use of chemical energy for charge storage based on electrostatic principle, which makes them complex to develop and use.

Supercapacitors: Ideal for short-duration high-power applications

Supercapacitors, also called ultracapacitors or electrochemical double-layer capacitors, have been the subject of focused R&D efforts for the past two decades. This has been facilitated by remarkable advances in material science and nanotechnology that allow construction of stacks of thin intercon-

nected electrochemical cells with very high surface-area electrodes which are generally made up of activated carbon, and low equivalent series resistors (ESRs). These devices can be manufactured with capacitance values of several hundred farads and even several thousand farads. Such values cannot be imagined in conventional capacitors. This translates into a new paradigm in energy storage, especially for short-duration, high-power applications. Depending on their design and construction, the specific power of supercapacitors can be ten to 100 times higher than of a battery.

Supercapacitors are mainly used for pulse power supply requirements. When you start the engine of an automatic-start vehicle, the starter motor draws a very large current from the lead-acid battery, which reduces the life of the battery. This high current requirement is due to high starting current of the starter motor. Such a high current is not required except for starting, which unnecessarily increases the battery size requirement. A supercapacitor can fulfil the require-

ment of high starting current, thereby reducing the load on the battery, increasing its life and reducing the battery size.

Leading car manufacturers such as Volvo, Opel, Rover, Renaults, Volkswagen and Daimler-Chrysler are using supercapacitors in their electric vehicles and electric hybrid vehicles. Regon's plot in Fig. 1 shows how supercapacitors (Boostcap by Maxwell) and lithium-ion batteries are approaching top right of the graph. If this combination achieves power and energy density of the internal combustion engine, your dependency on the IC engine and hence fossil fuels will reduce. Various electrical energy